

12

Liquidity, Price Discovery, and Corporate Policies

Learning Objectives:

- Effects of the financial markets' liquidity on firms' investment and corporate governance
- Effects of price discovery on corporate investment and executive compensation
- How corporate policies affect market liquidity

12.1 Introduction

A key function of securities markets is to channel funds from savers to firms, to finance their investment. What role do market liquidity and price discovery play in this process? For instance, is a more liquid market conducive to more investment? Does it lead to a better allocation of funds among alternative investment projects? These are old questions to which the great economists of the past have responded in very different ways.

John Hicks, for instance, argued that the liquidity of securities markets—and not technological progress per se—is what made the industrial revolution possible (Hicks 1969): “According to Hicks, the products manufactured during the first decades of the industrial revolution had been invented much earlier. Thus, technological innovation did not spark sustained growth. Many of the existing innovations, however, required large injections and long-run commitments of capital. The critical new ingredient that ignited growth in eighteenth century England was capital market liquidity” (Levine 1997, p. 692).

Keynes, by contrast, argued that liquidity can lead to inefficient investment decisions, by encouraging short-term speculation at the expense of sound investment decisions based on firms' long-term prospects: “If I am allowed to appropriate the term *speculation* for the activity of forecasting the psychology of the market, and the term *enterprise* for the activity of forecasting the prospective yield of assets over their whole life, it is by no means always the case that

speculation predominates over enterprise. As the organization of investment markets improves, the risk of the predominance of speculation does however increase. These tendencies are a scarcely avoidable outcome of our having successfully organized ‘liquid’ investment markets. It is usually agreed that casinos should, in the public interest, be inaccessible and expensive. And perhaps the same is true of Stock Exchanges” (Keynes 1936, pp. 158–9).

While these views conflict, they nevertheless both rest on the idea that liquidity is of paramount importance to firms’ investment, and therefore ultimately to their performance. In Section 12.2, we go deeper into the mechanisms through which liquidity affects firms’ investment decisions and discuss the evidence that has built up. Section 12.3 considers how high liquidity affects the governance of firms, and specifically whether it discourages the formation of large shareholding stakes and therefore the probability that management will be monitored by a large shareholder. But liquidity is not the only aspect of market microstructure that can affect corporate policies: the market’s ability to keep prices in line with fundamentals may also be important for firms. So Section 12.4 investigates the link between the accuracy of price discovery, the quality of firms’ investment decisions, and their ability to incentivize managers by indexing executives’ compensation to stock prices. Finally, Section 12.5 considers the reverse causal link, which is to say, actions that firms themselves can take to fine-tune the liquidity of their securities.

12.2 Market Liquidity and Corporate Investment

Chapter 10 provides an important reason why liquidity should enhance investment: investors will pay a higher price for more liquid securities or else, to the same effect, require a lower expected rate of return to hold them. Hence, a more liquid securities market lowers the cost of capital to firms, which should boost investment, as Hicks argued with reference to the industrial revolution: cheaper capital will increase the number and size of the investment projects with positive net present value, so more liquid markets should be associated with more investment and higher firm valuation.

There is evidence to support these predictions. Levine and Zervos (1998a), with data for 49 countries from 1976 to 1993, find that investment and economic growth are positively correlated with measures of stock market turnover, taken to be a proxy for liquidity. They estimate cross-sectional regressions where the dependent variable for a country is alternatively the investment rate, the growth rate of real per capita GDP, or the growth rate of productivity, averaged over the sample period. The explanatory variables include measures of stock market turnover in 1976, scaled by stock market capitalization or by GDP, plus several

control variables capturing other factors associated with capital accumulation and growth (initial income per capita; education; political stability; indicators of exchange rate and trade, fiscal, and monetary policy). The estimated coefficient of stock market turnover is always positive and significant, implying that stock market liquidity is positively and significantly correlated with subsequent investment.

Related evidence is drawn from the analysis of stock market liberalizations via policies relaxing restrictions on foreign investors' share purchases. These policies are associated with an increase in liquidity, a jump in stock prices, and a drop in the cost of equity capital, as well as an increase in private investment. For instance, in a sample of 11 developing countries that liberalized their stock markets, Henry (2000b) finds that the growth rate of private investment rose above the median pre-liberalization investment rate one year later in 9 countries, two years later in 10, and three years later in 8. The average growth rate of private investment in the three years after liberalization exceeds the mean of Henry's sample by 22 percentage points (see Section 12.6 for related studies).

There is also firm-level evidence that liquidity is correlated with company value. Fang, Noe, and Tice (2009) report that firms with more liquid stocks have higher market-to-book value ratios. To identify the causal effect of liquidity on firm performance, they focus on an exogenous shock to liquidity—the decrease of the tick size for U.S. securities in 2000 (see Chapter 6)—and show that the resulting increase in liquidity around that year raises company values. One mechanism that may explain this finding, as explained above, is the decline in firms' cost of capital due to better liquidity. An alternative mechanism is that a more liquid market may induce managers to select better investment opportunities, as the next section explains. Fang, Noe, and Tice (2009) try to discriminate empirically between these alternative mechanisms, and conclude that the cost-of-capital channel plays a minor role compared with the managerial decision channel.

Stock market liquidity is particularly important for the firms that first approach public markets via an IPO: their value is very uncertain, so the stocks of recent IPOs are particularly exposed to problems of asymmetric information. As information about the company progressively emerges in secondary market trading, the problem becomes less severe and market liquidity improves. Ellul and Pagano (2006) show that in a sample of 337 British IPOs between 1998 and 2000, the average bid-ask spread declined from 4.5% in the immediate aftermarket to about 2% several weeks after, as shown by Figure 12.1—a decline that largely reflects the decrease in the adverse-selection component. The figure also shows that the range of variation of the spread is greatest immediately after the IPO and declines steadily thereafter. This observation indicates not only that illiquidity is most severe in conjunction with the IPO, but that illiquidity risk

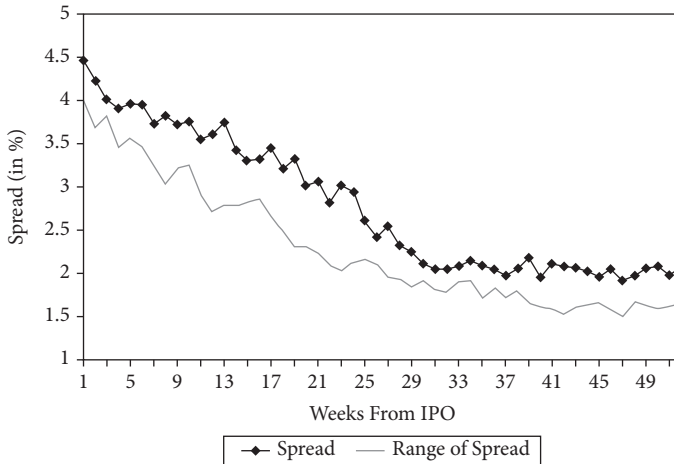


Figure 12.1 Bid-ask spread after an IPO: average value and range of variation (based on data from Ellul and Pagano 2006).

is too. Accordingly, the illiquidity premium that investors require to buy new stocks should be particularly large, especially if these stocks are purchased by “flippers” who want to buy and quickly resell.

Consistent with this argument, IPOs sell at a discount relative to the immediate post-IPO market price, known as “IPO underpricing”: relative to the first day of trading in the sample, this underpricing is 47.7%, which implies that firm owners leave a lot of money on the table when their firm goes public. In fact, the underpricing is more pronounced for firms that turn out to be more illiquid, as shown by Figure 12.2. In other words, the more illiquid the secondary market, the higher the cost of equity capital in the primary market for the companies that tap it. Ellul and Pagano (2006) show that this relationship persists even after controlling for other determinants of IPO underpricing, such as the reputation of the underwriter.

Another benefit of a liquid market is that it provides an “exit option” for venture capitalists—a class of intermediaries who invest in startup firms, acting not only as financiers but also as advisors and monitors in the early stages. As these advisory and monitoring services require highly specialized human capital, venture capitalists typically must limit their operations to just a few companies at a time. An increase in market liquidity enables them to exit more rapidly from their investments, because better prices are available to them. Michelacci and Suarez (2004) show that this more quickly frees their scarce human capital for redeployment in a new round of financing, and thereby increases the total number of firms they are able to serve.

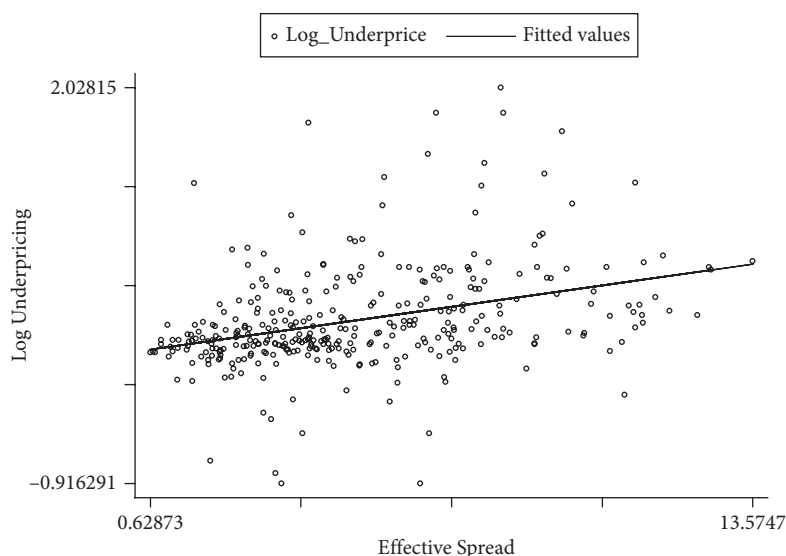


Figure 12.2 IPO underpricing and effective bid-ask spread after an IPO (based on data from Ellul and Pagano 2006).

12.3 Market Liquidity and Corporate Governance

We now turn to a more indirect channel through which liquidity affects firms' investment policies, namely via their corporate governance. In particular, the question is whether greater liquidity encourages investors to monitor companies in order to improve managerial decisions. As Section 10.1 indicates, Keynes would have probably answered in the negative. He viewed liquidity as discouraging investors from emphasizing the long-term prospects of their investments, as they can liquidate their holdings quickly and cheaply if dissatisfied with their performance. In line with this view, some have argued that the U.S. market-friendly regulatory framework, designed to enhance liquidity, ultimately discourages the activism of large shareholders.

Coffee (1991) states this view as follows: "Liquidity and control are antithetical. American law has said clearly and consistently since at least the 1930s that those who exercise control should not enjoy liquidity and vice versa. [...] the separation of liquidity and control is not only a cause of institutional passivity, but to some degree should be. In short, those institutions that most desire liquidity would make poor monitors." He adds that other countries have chosen a different point along the trade-off between liquidity and control: "Advanced industrial economies can be classified along a continuum ranging from those, such as Japan and Germany, that permit financial institutions to

control corporate managements, but effectively deny them liquidity, to those that inhibit institutional control, but maximize their liquidity.” This is represented by the United States, where disgruntled investors sell their shares rather than challenge management—the well-known “Wall Street Rule.” In this vein, Bhidé (1993) argues that in the U.S. “public policy has favored stock market liquidity over active investing,” at the cost of impaired corporate governance.

The trade-off between liquidity and control described by Coffee (1991) and Bhidé (1993) is a special case of A. O. Hirschman’s (1970) idea that members of a dysfunctional organization choose between exit and voice. If it is easy and cheap to exit by selling, shareholders will have little interest in exercising a costly voice to intervene actively in the affairs of the company, replacing unsatisfactory managers, and so on. But if exit is blocked because the market is illiquid, then exercising one’s “voice” becomes more attractive.

To capture these ideas, consider the time line of events illustrated by Figure 12.3: at date 0, a potentially activist shareholder amasses a block of shares amounting to a fraction ϕ of the firm’s equity, which has current value V ; at date 1 the blockholder decides whether to use his control rights to intervene in the company’s operation (voice) or to sell his stake (exit). We first analyze the date-1 problem facing the investor after he has already purchased a sizeable block of shares. Then we consider whether, given what he expects to happen at date 1, he wants to buy a block of shares at date 0.

If the blockholder intervenes at date 1, say by replacing lazy or incompetent management, he brings about a gain G in the firm’s value. But, to do so he must exert effort at a cost C . His decision whether to intervene or to walk away depends on the liquidity of the stock at date 1. If he stays for the long haul and intervenes, the value of his stake, net of the intervention cost, will be:

$$\phi(V + G) - C.$$

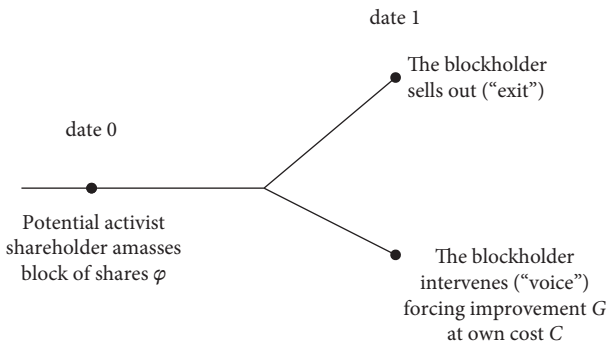


Figure 12.3 Stock market trading and shareholder activism: time line.

To make the problem interesting, we assume that the blockholder's intervention produces an increase in the value of his stake that exceeds the cost of intervention: $\varphi G > C$. Otherwise, there is no chance of activism.

If at date 1 the blockholder sells his stake, he gets:

$$\varphi(\mu_1 - S_1),$$

where μ_1 is the market's perception of the stock's value at date 1, including any downward revision induced by the blockholder's decision to sell, and the half-spread S_1 measures any market illiquidity over and above such price impact. For example, if trading is anonymous and the market believes the blockholder will intervene with probability π , then $\mu_1 = V + \pi G$. If instead regulation forces him to disclose his intention to exit before trading, then $\mu_1 = V$, because the market anticipates no improvement.

Thus, for voice to prevail over exit—i.e., for the shareholder to intervene rather than sell—we must have:

$$\varphi(V + G) - C \geq \varphi(\mu_1 - S_1). \quad (12.1)$$

Let us consider the case most favorable to exit, namely that the blockholder can sell anonymously at a price $\mu_1 = V + \pi G$, where π is a rational estimate of the probability of intervention. Then the previous condition (12.1) becomes:

$$\varphi G - C \geq \varphi(\pi G - S_1). \quad (12.2)$$

Clearly, this condition is always met if it holds for $\pi = 1$, and if so, there will always be intervention by the blockholder. Simple algebra shows that this occurs if $S_1 \geq C/\varphi$, that is, if the market is so illiquid that exit is excessively costly. If $S_1 < C/\varphi$ instead, there is a mixed-strategy equilibrium with the probability of intervention π such that the blockholder is just indifferent between voice and exit. In that case, condition (12.2) holds with equality. Encompassing both cases, the probability of intervention is

$$\pi = \min \left\{ 1 - \frac{1}{G} \left(\frac{C}{\varphi} - S_1 \right), 1 \right\}. \quad (12.3)$$

This expression shows that market illiquidity encourages voice by making exit more costly: the probability π of blockholder intervention increases with the illiquidity parameter S_1 , rising to 1 when $S_1 \geq C/\varphi$, as shown above.¹ This

¹ By assumption, $G - C/\varphi > 0 > -S_1$, so intervention will always occur with strict positive probability ($\pi > 0$), from equation (12.3).

illustrates the “lock-in effect” described by Coffee: the more illiquid the market, the more advantageous it is to monitor actively rather than to exit. Note also that blockholder activism is more likely, the greater the potential gain G , the lower the intervention cost C , and the larger the blockholder’s stake, as one would expect.

The lock-in effect just described is heightened if the blockholder is forced to disclose that he is walking away from his stake, so that the market prices the company at non-intervention value $\mu_1 = V$. In this case, exit becomes prohibitively expensive because there is no capital gain from selling, but the trading cost S_1 must still be paid. We leave it to the reader to verify that in this case activism is guaranteed, that is, $\pi = 1$ (see exercise 1).

Up to this point, we have taken the presence of a blockholder for granted. But we must consider whether a potential activist would find it worthwhile to build up a block of shares in the first place. As Bolton and von Thadden (1998) and Maug (1998) point out, the decision to purchase a block is affected by the liquidity of the market at the date of purchase: here liquidity favors activism by reducing the cost of building up the block. Formally, at date 0 the per-share price paid by the potential activist is $\mu_0 + S_0$, where μ_0 denotes the market’s valuation of the company and S_0 is the transaction cost. The market valuation μ_0 will range from V when the market does not expect a blockholder to emerge, to $V + \pi G$ when it does expect so (with probability π).

As before, consider the case where a blockholder’s presence is not foreseen, so that $\mu_0 = V$. Recall that, from the left-hand side of expression (12.1), the net value of a block after intervention at date 1 is $\varphi(V + G) - C$. Since the cost of amassing the block at date 0 is $\varphi(V + S_0)$, the ex ante net gain from doing so is:

$$\begin{aligned} & \varphi(V + G) - C - \varphi(V + S_0) \\ &= \varphi(G - S_0) - C, \end{aligned}$$

which is decreasing in the market illiquidity parameter S_0 . Hence illiquidity is a double-edged sword: it discourages the original formation of a block, but once the block is in place, it locks the blockholder in and encourages activism.

However, if at date 0 the market fully anticipates the presence of a blockholder, then any improvements that he may bring to the company will already be reflected in the date-0 share valuation $\mu_0 = V + \pi G$. In this case, he is certain to lose money if he goes ahead and acquires the block. To see this, note that his net gain is then:

$$\begin{aligned} & \varphi(V + G) - C - \varphi(V + \pi G + S_0) \\ &= \varphi((1 - \pi)G - S_0) - C, \end{aligned}$$

which, inserting the probability of his date-1 intervention from equation (12.3), becomes:

$$-\varphi S_0 - \min(C, \varphi S_1) < 0.$$

Intuitively, the blockholder incurs a transaction cost of S_0 per share when acquiring his stake, and subsequently incurs either the monitoring cost C or the exit transaction cost of S_1 per share, whichever is lower. Any improvements he makes to the company's value are already incorporated into the share price at purchase. Not surprisingly, he loses money on the round-trip transaction and so prefers to avoid it. This idea was first articulated by Grossman and Hart (1980), who argue that other shareholders free-ride on the value improvements they expect the activist investor to make, by refusing to sell at any price that does not fully reflect them. Thus, the potential activist cannot recover his costs, unless he can secretly build up a toehold stake at a lower price. This will largely depend on his ability to camouflage his trades behind noise trading, as Kyle and Vila (1991) note.

The model presented above indicates that the assertion that "liquidity and control are antithetical" by Coffee (1991) is an oversimplification: greater liquidity may actually facilitate block formation, and therefore active intervention in corporate control by blockholders.² The relationship between liquidity and activism is further complicated by the fact that, from the standpoint of a large shareholder, liquidity will also depend on the price impact of his trades, in addition to S_0 and S_1 , which may discourage the accumulation of a large stake and penalize its decumulation. As noted, the magnitude of this adverse price impact depends on the visibility of the blockholder's trades. If regulators want to promote the formation of activist blockholders, they will have to facilitate accumulation of significant blocks of shares without disclosure to other market participants. But once a blockholder does have a controlling stake, sale of this stake should be highly visible to make his exit as costly as possible. In other words, the formation of a block requires relative opacity, while its retention requires great transparency, a delicate balancing act for regulators.

² Edmans (2009) points out an additional reason why liquidity and monitoring by large shareholders are complementary, rather than antithetical. Since in a liquid market a large shareholder has low exit costs after bad investment decisions, his trading decisions can convey more information to the market about the quality of managerial decisions and so reinforce the discipline to which managers are subject. By the same token, liquidity enhances the signal conveyed by the blockholder's retention of his stake when the firm books low earnings: precisely because he could sell at little cost, his decision not to do so signals that the fundamental value of the firm is sound, and keeps the stock price well aligned with fundamentals. Edmans also shows that when the blockholder chooses his stake optimally, greater market liquidity induces the blockholder to choose a larger stake, which makes the stock price more informative about the long-run payoff of the firm and induces management to avoid short-termist investment behavior.

In practice, regulators have addressed this dilemma by allowing the buildup of an initial stake to go undisclosed below some threshold (in the United States, 5% of a firm's equity); beyond it, all further transactions by the blockholder must be disclosed immediately. Thus, once the stake is taken, it becomes hard to decumulate it without a price concession that reflects the lower monitoring incentive by the blockholder.

Empirically, the presence of blockholders varies greatly from country to country. Blocks are commonplace in continental Europe and Asia but much less so in the United States and United Kingdom. Even in the United States, however, the involvement of large institutional shareholders increased dramatically with the advent of public pension fund activism during the mid-1980s and the greater role of mutual funds in equity investing. Davis and Yoo (2003) report that mutual funds manage over 20% of U.S. equity and hold sizeable blocks of 10% or more in many of the largest U.S. companies.

There is evidence that institutional investors do regard corporate governance as important to their investment decisions, but nevertheless tend to exit rather than voice their discontent (see Section 12.6 for details). This aversion to exerting influence may be due partly to the U.S. short-swing profit rule, which obliges investors who actively participate in management decisions (or hold a stake of at least 5%) to hold their stake for at least six months, or else disgorge any capital gains made on the in-and-out transaction. This deters institutions like mutual funds from activism, as they may experience unpredictable outflows that oblige them to liquidate shares.

Another reason for the lack of activism of institutional investors—even a frequent pro-management attitude—is conflict of interest: not opposing or even backing the managers of the companies in which they invest may enable them to get or keep lucrative contracts to run those companies' pension funds. Davis and Kim (2007) analyze proxy voting by U.S. mutual funds in 2004 and find that aggregate pro-management votes at the fund family level are positively correlated with the magnitude of mutual funds' business ties to firms in their portfolio, even though the correlation does not emerge when examining voting behavior at specific firms: in other words, funds with strong business ties with portfolio firms tend to be less activist across the board; they do not necessarily cast their votes specifically in favor of client firm management.

A more fundamental question is whether shareholder activism actually prompts better corporate performance. The evidence is mixed. While some studies document positive short-term market price reactions to announcements of certain kinds of activism, others find that large blockholders fail to improve the long-term operating or stock market performance of the targeted companies (for a survey, see Gillan and Starks 2007). One reason for this mixed evidence

may be that often activist shareholders intervene “behind the scenes,” so their actions are not picked up by studies that rely on public information. Indeed Becht et al. (2009) document the effectiveness of such behind-the-scenes activity, using data from British Telecom’s pension fund manager Hermes on its private engagements with management in companies targeted by its U.K. Focus Fund. They find that the fund intervenes predominantly through such private engagements and, in contrast with most previous studies of activism, they report that the Hermes fund considerably outperforms benchmarks and that its abnormally high returns are correlated with its interventions. McCahery, Sautner, and Starks (2016) provide survey-based evidence that these behind-the-scenes practices by institutional investors are widespread, and are used jointly with governance-motivated exit. These governance mechanisms are viewed as complementary devices, with intervention typically occurring prior to a potential exit. They also find that long-term investors and investors less concerned about stock liquidity intervene more intensively. Becht, Franks, and Wagner (2021) provide further evidence that the monitoring effort of active asset managers may lead to their exit: they show that a large institutional investor that engages in high-level private meetings with executives and board members of portfolio firms gains a sizeable information advantage that it exploits in trading on and around meeting days and votes.

There is more agreement on the evidence regarding shareholder activism by hedge funds. These are more lightly regulated and have fewer conflicts of interests than other institutional investors. Most empirical studies support the view that their activism creates value for shareholders by improving the governance, capital structure decisions, and operating performance of target firms (for a survey, see Brav, Jiang, and Kim 2009). Brav, Jiang, Thomas, and Partnoy (2008), using a large data set from 2001–6, find that activist hedge funds in the United States attain success or partial success in two-thirds of the cases in which they intervene, and that the announcement of their activism is accompanied by an abnormal return of approximately 7%, with no reversal during the subsequent year, and is followed by higher payout, better operating performance, and greater CEO turnover.

Interestingly, Edmans, Fang, and Zur (2013) find that market liquidity increases hedge funds’ propensity to acquire large blocks of shares, but that once the stake is acquired, liquidity increases the likelihood of hedge funds opting for “exit” rather than voice, exactly as predicted by the model here: greater liquidity makes it more likely that the blockholder will choose a Schedule 13Gs filing (passive investment) rather than 13Ds (active investment), especially if its manager’s wealth is sensitive to the stock price.

12.4 Price Discovery, Corporate Investment, and Executive Compensation

Sections 12.2 and 12.3 examined how market liquidity affects two key aspects of corporate policy, namely investment and governance. But these decisions are also affected by price discovery. An essential function of securities markets is discovering asset values by aggregating investors' private signals about future cash flows. The information conveyed by market prices can then be used by managers to guide their investment decisions; this is the topic of Section 12.4.1. If stock prices provide information about a company's performance, they can also be used by shareholders to index compensation and give executives sharper incentives to maximize the company's value, as will be discussed in Section 12.4.2.

12.4.1 Stock Prices and Investment Allocation

The information conveyed by stock market prices can be useful to firms' managers and stakeholders more generally. For instance, managers may decide to pursue or terminate an investment plan (e.g., a major acquisition, a research and development project, or diversification into new products and markets) after observing the market's reaction to its announcement. In line with this idea, Luo (2005) studies the case of merger announcements empirically and shows that managers use the stock price reaction to these announcements in deciding whether to cancel or consummate the deal. Or a sharp decline in stock prices may suggest to a financier that other investors have adverse information on the firm's prospects and induce him raise the return required for new capital (Goldstein, Ozdenoren, and Yuan 2009).

Box 12.1 Strategy Changes and Stock Price Reactions: HP'S Double U-Turn

On August 19, 2011, the price of the computer company Hewlett Packard's (HP) stock plunged 20.5%, after a decline of 6% the previous day. This followed an announcement by HP's chief executive officer Leo Apotheker that he planned to spin off or sell the personal computer unit for which HP was universally known and refocus the company on software production with the planned \$11 billion acquisition of Autonomy, Britain's largest software producer. The new strategy was motivated by considerably larger profit

margins in software than in personal computers. Clearly, the stock market did not react well to the announcement. On September 22, the company fired Mr. Apotheker and replaced him with Meg Whitman, former CEO of eBay. On October 27, HP announced that it had dropped the plan to spin off the personal computer unit, largely because it would have lost synergies related to bulk purchasing of components, thus significantly increasing costs. The price of HP shares rose by 22.6% from \$22.80 on September 22 to \$27.95 on November 30. However, the Autonomy deal was not abandoned, and it later resulted in extensive litigation regarding misleading accounting practices at Autonomy in the runup to the deal; and in 2021, HP wrote down a record \$8.8 billion of Autonomy's value.

The possibility that managers' investment decisions may be determined by stock prices creates a complex interdependence between the stock market and the real economy, as first modeled by Dow and Gorton (1997). It is clear that real investment decisions affect stock prices, since the fundamental value of a security is the discounted value of its future cash flows. But if managers gather information from stock prices, causation may also run the other way. That is, stock prices will also affect investment decisions, hence future cash flows.

To capture this feedback from stock price to investment, consider the sequence of actions illustrated in Figure 12.4. At date 0, a firm announces that it is considering a new investment project (e.g., an R&D program for a new chip or a move into a new product line as in the example of Box 12.1), which is equally likely to be of high quality (H) or low quality (L). The project will contribute a net gain $G > 0$ to the company's value (originally equal to V) if it is of high quality or else entail the loss of the investment I :

$$\Delta V = \begin{cases} G & \text{if the company invests and project is high-quality (H),} \\ -I & \text{if the company invests and project is low-quality (L),} \\ 0 & \text{if the company does not invest.} \end{cases}$$



Figure 12.4 Stock prices and investment decisions: time line.

It is assumed that, if no information about quality is available, the project is not viable: $G < I$, so that the project's expected net present value, $\frac{1}{2}G - \frac{1}{2}I$, is negative.

At date 1, after the announcement of this investment opportunity, the firm's shares are traded by investors, some of whom have valuable information about the quality of the project. As in Chapter 3, two potential types of traders place orders that are filled by risk-neutral, competitive, and uninformed market makers. With probability π , an informed risk-neutral speculator who knows whether the project's quality is H or L comes to the market, and makes a trade that maximizes his expected profit. With probability $1 - \pi$, a liquidity trader buys or sells one unit of stock with equal probabilities. The market makers price the stock at its expected value, conditional on the arrival of a buy or sell order.

At the same time, with probability γ the firm's manager privately receives a report that enables him to establish the quality of the project; with probability $1 - \gamma$ he does not. Thus, at date 2, when the decision to proceed with the project or not must be taken, the manager considers both his own information (if any) and the reaction of the stock market to the announcement. The issue is whether and how the information conveyed by trades on the stock market can help guide the investment decision.

To solve for the equilibrium, let us conjecture that informed traders buy a unit of the stock if they learn that the project is of high quality and sell a unit otherwise: later we will check that this strategy is indeed optimal in equilibrium.

If the manager has gotten a private report, he needs look no further to decide whether to invest. The interesting case is when he has not received any such report, and so his only guide is the stock market. As in Chapter 3, if a buy order is placed, it executes at the ask a ; a sell order executes at the bid b . Hence the transaction price p reveals the direction of trading interest. Depending on whether he observes a transaction at ask or at bid, the manager updates his estimate of the probability of the project being of quality H , using Bayes' Law:³

$$\Pr(H|p = a) = \frac{1 + \pi}{2},$$

$$\Pr(H|p = b) = \frac{1 - \pi}{2}.$$

If the manager observes a trade at the ask, then his updated expectation of the project's net present value is:

³ These two probabilities can be obtained from equations (3.16) and (3.17), which describe the updating of the dealer's beliefs in Chapter 3. Setting the prior probability of the stock being high value $\theta_{i-1} = \frac{1}{2}$ yields the updated probabilities $\theta_i^+ = \Pr(H|p = a)$ and $\theta_i^- = \Pr(H|p = b)$.

$$\begin{aligned} & \Pr(H|p=a)G + (1 - \Pr(H|p=a))(-I) \\ &= \frac{1+\pi}{2}G - \frac{1-\pi}{2}I. \end{aligned} \quad (12.4)$$

A transaction at the ask will persuade him to invest if this updated expected present value is positive, which requires that:

$$\pi \geq \frac{I-G}{I+G}. \quad (12.5)$$

Recalling that π is the proportion of informed investors, inequality (12.5) states that the executive's decision is guided by the stock market only if the trading is informative enough. This inequality is more easily satisfied if I is very close to G , that is, if the project is borderline in the first place, so that even a little bit of positive news can tip the balance. Naturally, if a transaction is observed at the bid, the manager elects not to invest: his prior estimate of the project's net present value was already negative by assumption, and his posterior estimate will be even lower.

What is the allocative value of the information from the stock market? Let us compare the outcome of the investment decision with and without a stock market. We assume that the informativeness condition (12.5) holds, because otherwise the stock market would clearly play no role.

In the absence of a stock market (that is, if the company is not listed), the manager will only invest with probability $\gamma/2$, namely if he receives private information about the project and if the information is positive. Therefore, ex ante the value of the firm, if privately held, is:

$$V_{private} = V + \frac{\gamma}{2}G. \quad (12.6)$$

If the firm is listed, however, the manager can fine-tune his investment decision and invest even if he does not have private information, as long as he observes a trade at the ask price on the market. As before, he gets positive private information with probability $\gamma/2$, in which case he again invests gaining G . But when he does not receive private information, which occurs with probability $1-\gamma$, he will observe a transaction at the ask with probability $\frac{1}{2}$ and therefore invest. The net present value of this policy is given by equation (12.4), so that the ex ante value of the firm when publicly listed is:

$$\begin{aligned} V_{public} &= V + \frac{\gamma}{2}G + \frac{1-\gamma}{2} \left(\frac{1+\pi}{2}G - \frac{1-\pi}{2}I \right) \\ &= \underbrace{V + \frac{\gamma}{2}G}_{V_{private}} + \underbrace{(1-\gamma)\frac{\pi}{2}G}_{\text{informational gain}} - \underbrace{(1-\gamma)\frac{1-\pi}{2}\frac{I-G}{2}}_{\text{loss from noise}}. \end{aligned} \quad (12.7)$$

Thus we can think of the total contribution of the stock market to the investment decision as consisting of two components: the gain from investing when the informed trader is present and provides a valuable positive signal by buying the stock, and the loss from overinvesting when the noise trader buys stock (in which case the investment on average is mistaken, given our assumption that $I > G$). Clearly, the net effect is positive whenever the informativeness condition (12.5) is satisfied: the manager will only consider the stock price when it improves the firm's investment decision. And as one would expect, the net gain is increasing in the informativeness of trading, as measured by π .

Under our assumptions, the stock market encourages investment, insofar as it sometimes prompts the manager to invest even in the absence of private information. However, this feature of the model is not robust: if we were to assume that on average the investment is viable (i.e., if we assumed that $G > I$), the stock market would act as a brake on investment, by prompting the manager to forgo investment when traders sell. We leave this case as an exercise to the reader (see exercise 2).

Having obtained the manager's equilibrium investment strategy, we can now compute the bid and ask prices that will prevail on the market in equilibrium. This problem is somewhat more subtle here than in Chapter 3, because market makers must take into account the relationship between market trading, the manager's investment decision, and the probability of the project's being high quality. Recall that market makers have no independent information and simply condition their quotes on whether there is a buy or a sell order. Computing the equilibrium bid and ask prices is also useful to check the validity of our initial conjecture that the informed speculator always finds it profitable to buy when he learns that the company's project is high quality, and to sell if it is low quality.

Since equilibrium bid and ask prices depend on the investment policy, they differ depending on whether the informativeness condition (12.5) holds or not. If it does, market makers know that the manager will invest either when he receives positive private information or when he observes a buy order. So, competitive market makers will set their ask price at:

$$a = V + (1 - \pi) \underbrace{\left[\frac{\gamma}{2} G + (1 - \gamma) \frac{G - I}{2} \right]}_{\text{NPV if noise trader buys}} + \underbrace{\pi G}_{\substack{\text{NPV if} \\ \text{informed} \\ \text{buys}}} \quad (12.8)$$

The term in square brackets is the net present value of the investment conditional on the order coming from a noise trader: with probability $\gamma/2$, the manager knows that the project is high quality and invests; with probability $\gamma/2$, he knows that it is low quality and does not invest; and with probability $1 - \gamma$, he has no private information and mistakenly takes guidance from the stock market, so that the project's expected value is $(G - I)/2$ (since the noise trade contains

no real information). The last term in the expression is the value of the investment conditional on the buy order coming from an informed trader, namely G .

Conversely, a competitive market maker will set his bid price at:

$$b = V + (1 - \pi) \underbrace{\frac{\gamma}{2} G}_{\text{NPV if noise trader sells}} \quad (12.9)$$

If there is a sell order, the manager will only invest if he has positive private information. In this case, the sell order necessarily comes from a noise trader because the manager's information and the informed trader's signal never conflict. Hence, the market maker anticipates a successful investment (worth G) with probability $(1 - \pi)\gamma/2$ (the joint probability of a noise trade and positive private information).

How will the informed speculator react to the ask and bid prices in (12.8) and (12.9)? If he knows that the project is of high quality, and expects the manager to invest upon observing the speculator's own buy order, he will value the firm at $V + G$. Hence, by placing a buy order at the ask in (12.8) he will book the expected trading profit:

$$(V + G) - a = \frac{1 - \pi}{2} [G + (1 - \gamma)I] > 0. \quad (12.10)$$

Similarly, if he knows that the project is of low quality, the speculator will book a profit from a sell order at the bid price in (12.9). In this case, since the manager will not invest upon observing the speculator's own sell order, the firm will be worth V , so that a sell order implies the expected profit is:

$$b - V = (1 - \pi) \frac{\gamma}{2} G > 0. \quad (12.11)$$

It can also be readily shown that the informed speculator has no incentive to trade against his information (see exercise 3): for instance, he does not short the security when he knows that the investment is intrinsically sound.⁴

Hence, as initially conjectured, in equilibrium the speculator buys upon receiving good news about the project and sells otherwise. We leave it to the reader to show that the same applies if the informativeness condition (12.5)

⁴ This type of manipulation can happen, however, in more complex models with feedback from the stock market to investment decisions, as shown by Goldstein and Gümbel (2008). In this case, instead of contributing to the efficient allocation of corporate investment, the signals issued by stock market prices would impair it.

does not hold, so that the stock market does not affect the manager's investment decisions (see exercise 4).

To sum up, we have established that there is an equilibrium in which the informed speculator buys if he receives a positive signal about the manager's strategy but sells otherwise, and the manager implements his strategy if he receives good information about the project or if the stock market's reaction to the announcement of his strategy is positive, but not otherwise.

Using equations (12.8) and (12.9), one can compute the equilibrium bid-ask spread:

$$S = \pi G - \frac{(1-\pi)(1-\gamma)}{2}(I-G). \quad (12.12)$$

Clearly, S is increasing in π .⁵ So, as in Chapter 3, an increase in the prevalence of informed trading not only increases the informational content of stock prices but also makes the market less liquid. This points to an interesting conclusion: informed trading π increases the stock market's ability to guide investment, as measured by the gain $V_{\text{public}} - V_{\text{private}}$ in (12.7), at the cost of reducing market liquidity, as measured by the bid-ask spread (12.12). This highlights the danger of considering liquidity as a universally desirable attribute of the stock market, and provides a counterexample to the view that liquidity increases asset values and investment, as suggested by the analysis in Chapter 10. In this model, more frequent informed trading reduces liquidity, but on average raises the stock price, because it improves the allocation of investment, as is attested to by the fact that the average transaction price (the midprice, since buyers and sellers arrive with equal probabilities in this example) is increasing in π :

$$m = \frac{a+b}{2} = V + \frac{1}{2} \left\{ [(1-\gamma)\pi + \gamma]G - \frac{(1-\pi)(1-\gamma)}{2}(I-G) \right\}.$$

In this model, the frequency of informed trading also increases the frequency of investment: if π is so low that it does not satisfy condition (12.5), the manager invests only with positive private information, that is, with probability $\gamma/2$; if instead π is above the threshold set by condition (12.5), he invests also when he sees buying in the stock market, that is, with probability $\frac{1}{2}$.⁶

⁵ The equilibrium spread in (12.12) is positive, since we assume that condition (12.5) holds, that is, $\pi \geq \frac{I-G}{I+G}$ and, by the same token, $1-\pi \leq \frac{2G}{I+G}$. Using these two conditions for π and $1-\pi$ and equation (12.5), we find that $S \geq \gamma G(I-G)/(I+G)$, which is non-negative, so that $S \geq 0$ for any π that satisfies condition (12.5).

⁶ The argument developed so far has an important limitation: it takes the frequency of informed trading as a given parameter π , whereas in reality it could be endogenous. In markets where there is

We discuss evidence regarding this feedback from stock prices to investment decisions, via managerial learning from prices, in Section 12.6.

12.4.2 Stock Prices and Executive Compensation

A central theme in corporate finance is that, when control is entrusted to professional managers, the interests of shareholders and management may conflict. These conflicts can take many forms. Managers may pursue objectives that are not in the shareholders' best interest: for instance, they may opt for an "easy life" rather than work hard to seek the best profit opportunities for the company; they may pursue unwarranted expansion (empire building) or they may give themselves unduly large salaries or perks, or make nepotistic appointments. Alternatively, they may deviate from shareholder value maximization in the interest of other stakeholders, for example, undertaking loss-making but environmentally friendly projects or retaining unproductive employees. It can be difficult or inconvenient for shareholders to monitor and restrain such opportunism, especially if ownership is dispersed, so that shareholders' stakes are too small to justify the costs of activism.

And even if shareholders could verify managers' behavior, it may be very hard to curtail their opportunism. Consider an executive who is seen to opt for an "easy life," as opposed to hard work: it may be very difficult for the shareholders to take such a manager to court and prove that he did not work as hard as possible. The reason is that "working hard" cannot be accurately described in a contract enforceable in court: it is impossible to foresee all the contingencies that can arise in managing the firm and specify what a "hardworking" manager must do in each. Indeed, in the United States the so-called "business judgment rule" protects managers from legal challenges to their decisions (except in clear cases of negligence or corruption) and thus gives them considerable flexibility to act as they deem appropriate in the company's interest. In more technical jargon, even if shareholders could observe the manager's effort level, such effort may not be verifiable or legally enforceable. This generates a moral hazard (or agency) problem between shareholders and management.

While lawsuits may be ineffective, shareholders can mitigate this moral hazard and thus improve the company's performance by giving management financial

more trading by uninformed investors, speculators can obtain higher returns on their information, and so have more incentive to acquire information on the quality of firms' investment projects. Since more uninformed trading also tends to increase liquidity (as in Kyle's model), changes in it may result in a positive correlation between liquidity and the frequency of informed trading, in contrast with the tradeoff between liquidity and the allocative value of prices described in the text.

incentives to work hard.⁷ By indexing the manager's compensation to some measure of company performance (based on accounting variables or on stock prices), shareholders can induce the manager to exert effort—a practice known as “pay for performance.” Holmstrom and Tirole (1993) contend that linking executive pay to the price of their company's stock may be very effective in sharpening their incentives to exert effort. They argue that if shares are listed and publicly traded, speculators will gather information on the firm's performance and the stock price will thus at least partly reflect this information. The more precise the information conveyed by the firm's stock price, the more effective linking managerial compensation to it will be in eliciting effort, and hence the greater the enhancement of the firm's performance.

In this section, we capture this point in a simple setting where shareholders can choose whether the manager's compensation is to depend on the final value of the company (say, its long-term earnings or its terminal liquidation value) or to the price of the stock at some interim date, when investors observe the effort exerted. Specifically, as illustrated in Figure 12.5, at date 0 shareholders design the compensation contract; at date 1 the manager chooses whether to exert effort at a personal cost c (a choice that is publicly observed, though not verifiable in court); at date 2, the shares are traded on the stock market at price P ; finally, at date 3, the final value of the company V —say, its earnings—is either high (V^H) or low (V^L). The manager's effort increases the company's expected value: the

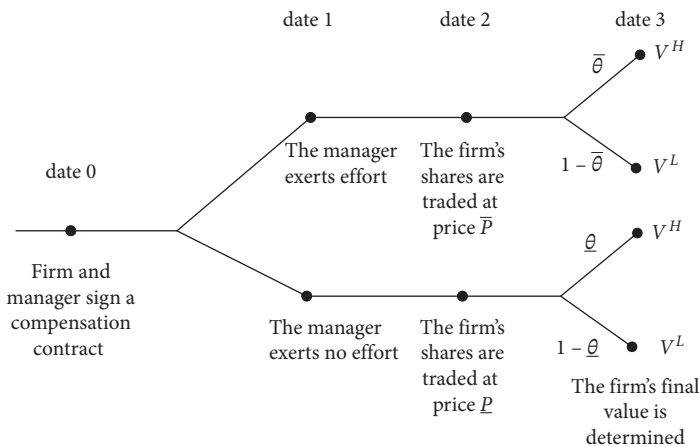


Figure 12.5 Stock prices and managerial compensation: time line.

⁷ Other governance mechanisms that can realign the executive's incentives with those of shareholders are the risk of termination (i.e., managerial turnover) or career concerns (performance-based advancement).

probability that $V = V^H$ is $\underline{\theta}$ if the manager exerts no effort, $\bar{\theta} = \underline{\theta} + \Delta\theta$ if he exerts effort.

Thus, if he exerts effort, the manager increases the company's expected value by $\Delta\theta(V^H - V^L) = \Delta\theta\Delta V$. We assume that effort is efficient, i.e., that the effect on the company's expected final value is greater than the cost of the manager's effort: $\Delta V > c/\Delta\theta$. In other words, shareholders would find it worthwhile to just compensate the manager for his effort if they could make sure that the manager does not shirk. For simplicity, we assume that the manager's reservation wage (i.e., what he could get from another employer) is zero, so he will be willing to work for the firm if his expected wage from exceeds his effort cost, that is, his participation constraint is satisfied. Moreover, the manager is assumed to enjoy limited liability: he cannot be made to pay any penalty over and above his compensation. Hence, his wage can never be negative.

Consider first the case in which the manager's pay can only be conditioned on the final value of the company (i.e., incentive pay can only be based on reported earnings: pay $w(v)$ takes one of two values w^H or w^L depending on the firm's final value). It is most economical for the owners to set the salary as low as possible in case of failure: $w^L = 0$. Incentive compatibility requires that the manager be given a high enough compensation w^H in case of success to induce him to exert effort:

$$\bar{\theta}w^H - c \geq \underline{\theta}w^H. \quad (12.13)$$

The lowest-cost pay scheme that elicits effort is thus:

$$w(v) = \begin{cases} w^H = c/\Delta\theta & \text{if } V = V^H, \\ w^L = 0 & \text{if } V = V^L. \end{cases} \quad (12.14)$$

The manager's participation constraint is satisfied, since this compensation package gives him a positive expected payoff $\bar{\theta}w^H - c = \underline{\theta}c/\Delta\theta > 0$, which is greater than his zero reservation wage, an example of what is known in the labor economics literature as an "efficiency wage," surveyed by Yellen (1984). So shareholders end up paying the manager a "rent" to stimulate him to exert effort. This is due to the manager's limited liability: if shareholders could inflict a penalty when the company's value is low, they could avoid paying him this rent, since they could incentivize him while just meeting his participation constraint.⁸ Hence the company's expected value, at time 0, is:

⁸ Inflicting a penalty on an underperforming manager is equivalent to paying him a negative salary. If shareholders could pay a negative salary $w^L < 0$ to a manager for poor performance, they could meet his incentive constraint by paying him $w^H = c/\Delta\theta + w^L$ if $V = V^H$ and w^L if $V = V^L$. But now they can also reduce his wages to just barely meet his participation constraint by setting his average wage, net of effort cost, to zero: $\bar{\theta}w^H + (1 - \bar{\theta})w^L = c$, so as to just compensate him for

$$V_0 = [\bar{\theta}V^H + (1 - \bar{\theta})V^L] - \bar{\theta}w^H = V^L + \bar{\theta}\left(\Delta V - \frac{c}{\Delta\theta}\right). \quad (12.15)$$

So far, it has been assumed that, although the company's shares are publicly traded, shareholders do not take the stock price into account in determining the manager's pay. But, at date 2, the stock price will be higher when the manager exerts effort than when he does not since investors observe the manager's effort.⁹ Suppose now that the manager's compensation can be conditioned on the stock price rise associated with his effort: he can be paid a salary $\bar{w}$ if the stock price signals that the manager exerted effort at date 1, and zero otherwise. Hence, at date 2, the company's stock price is:

$$P = \begin{cases} \bar{P} = \underline{\theta}V^H + (1 - \underline{\theta})V^L + \Delta\theta\Delta V - \bar{w} & \text{if the manager exerts effort,} \\ \underline{P} = \underline{\theta}V^H + (1 - \underline{\theta})V^L & \text{if the manager shirks.} \end{cases} \quad (12.16)$$

The incentive constraint now is:

$$\bar{w} - c \geq 0. \quad (12.17)$$

Hence, shareholders optimally offer the following compensation package to the manager:

$$w(P) = \begin{cases} \bar{w} = c & \text{if } P = \bar{P}, \\ \underline{w} = 0 & \text{if } P = \underline{P}. \end{cases} \quad (12.18)$$

Under this compensation scheme, from expression (12.16), $\bar{P} - \underline{P} = \Delta\theta\Delta V - c$. This difference is positive: as conjectured, the company's stock price is higher when the manager is seen to exert effort ($\bar{P} > \underline{P}$).

The key point is that, by linking the manager's pay to the stock's performance, shareholders can incentivize the manager without leaving any rent to him: they just compensate him for his effort, so his net payoff is zero. The reason is that in this model the stock price is a more precise (in fact, here perfect) signal of the effort than reported earnings. As a result, with stock-based compensation the expected value of the company at date 0 will be greater than under earnings-based compensation:

the cost of effort. Hence, the manager's optimal compensation would be $w^H = c(1 - \bar{\theta})/\Delta\theta > 0$ if $V = V^H$ and $w^L = -c\bar{\theta}/\Delta\theta < 0$ if $V = V^L$.

⁹ The difference is $\Delta\theta\Delta V - c > 0$. The proof is left to the reader as an exercise.

$$V'_0 = [\bar{\theta} V^H + (1 - \bar{\theta}) V^L] - \bar{w} = V^L + (\bar{\theta} \Delta V - c), \quad (12.19)$$

which exceeds the value in equation (12.15). The gain in value is $V'_0 - V_0 = c\bar{\theta}/\Delta\theta$. Thus the increase in expected value is proportional to $\bar{\theta}$, the likelihood that the company will overperform when the manager shirks, and to c , the private cost of effort to the manager. Intuitively, a higher value of either one of these parameters worsens the problem of moral hazard and so increases the desirability of indexing the manager's compensation to stock prices. In exercise 6, the reader is invited to solve for the optimal compensation scheme when investors do not necessarily observe the manager's effort.

While this model shows that stock-based compensation can benefit shareholders by sharpening managerial incentives, this type of scheme also has some drawbacks. One is pointed out by Holmstrom and Tirole (1993) themselves: stock prices are a reliable device for managerial discipline only if they are informative about the company's future performance, but the greater informativeness of stock prices comes at the cost of lower liquidity, because the informational rents of speculators are gained at the expense of uninformed investors. If these investors are rational, initially, they will be willing to buy the company's shares only at a lower price—they will require an illiquidity discount.

As shown in Section 10.3, more informed trading raises this illiquidity premium and so lowers the initial price that the security will command. This factor works in the opposite direction from that considered so far: while stock price informativeness tends to increase the company's initial value by enabling shareholders to better discipline managers, the related reduction in market liquidity tends to reduce value. The familiar tension between price discovery and market liquidity resurfaces. Holmstrom and Tirole (1993) conclude that companies will have to trade off the beneficial impact of the increase in executive effort against the implied illiquidity discount.

Another disadvantage of stock-based (and option-based) managerial compensation is that it may lead managers to divert valuable resources to misrepresent performance (Goldman and Sleazak 2006). This danger has been shown to be empirically relevant by a number of studies, which detect a positive correlation between managerial incentive pay and accounting fraud (see Section 12.6). Moreover, stock- and option-based compensation may induce managers to pursue short-term rather than long-term objectives: for instance, Edmans, Fang, and Lewellen (2017) identify short-termism using the amount of stock and options scheduled to vest in a given quarter, and find that vesting equity is associated with a decline in the growth of research and development and capital expenditure in the same quarter. Last but not least, stock- and option-based compensation has been shown to be associated with excess risk-taking

by bank managers, especially in the context of the 2007–9 financial crisis (see Section 12.6).

12.5 Corporate Policies and Market Liquidity

We have seen that a liquid security market can increase corporate investment and that the informational content of stock prices can improve firms' investment decisions and help them to provide incentives to management. This suggests that companies may wish to enhance the liquidity of their stock or the informativeness of its price. And companies can in fact affect these variables by a variety of decisions.

Clearly, the most basic decision in this respect is whether or not to go public. But even after this decision is made, this is not the end of the story: after an IPO on its domestic stock market, a company may decide to cross-list on other exchanges. Alternatively, it can choose from the start to list on multiple exchanges, via a global IPO. The determinants and effects of these choices are discussed briefly in Section 12.5.1.

Public listing, however, is no guarantee that the shares will be actively traded, with narrow bid-ask spreads and informative prices. To achieve this, a company can act on three other fronts. First, it can support the liquidity of its shares directly by paying designated market makers to trade in its stock and maintain a tight bid-ask spread: Section 12.5.2 discusses the effects of these policies on liquidity and company value. Second, the company can affect the liquidity of its securities by disclosing detailed, precise and timely value-relevant accounting data and by encouraging financial analysts to evaluate and convey them to investors, as discussed in Section 12.5.3. Third, it can affect liquidity and informativeness by adjusting its issuance of shares and debt securities, hence its leverage—the key parameter of a firm's capital structure. Alternatively, it can tweak the design of its securities so as to vary the sensitivity of the security's price to performance information. These issues are discussed in Section 12.5.4.

12.5.1 Listing and Cross-Listing

Liquidity is the most obvious benefit from listing a company's shares on a public market: the trading platform for its stock acts as a coordination device, where potential buyers and sellers can meet and trade, instead of searching informally for a counterparty. Per se, this liquidity gain expands the company's shareownership base to investors who would otherwise shun it, for fear of being

unable to liquidate the shares at a reasonable price in case of need. In fact, many investors may not even be aware of a company's existence until it lists. As highlighted by Merton (1987), investors' portfolio choices are constrained by their limited awareness of which securities are available for investment.

Hence, going public should enable companies not only to reduce the cost of equity financing—thanks to the lower liquidity premium (see Chapter 10)—but also to access a larger pool of external funds, by expanding their shareholder base. These effects are further amplified by the fact that public companies typically disclose much more information than private ones do (Section 12.5.3). In the United States, young companies tend to use their improved post-IPO access to external equity to increase investment, while more mature companies use it mainly to pay down debt (Mikkelsen, Partch, and Shah 1997); in Italy, instead, newly listed companies use the funds mainly to reduce leverage, and negotiate lower interest rates with their banks (Pagano, Panetta, and Zingales 1998).

The benefits of going public extend beyond liquidity and access to finance. An IPO enables the controlling shareholders to divest some of their shares and diversify their portfolios (Pagano 1993b; Chemmanur and Fulghieri 1999). It also allows the company to use the stock price as an input into managerial decisions (Section 12.4.1) and can improve output market performance by enhancing the firm's visibility and reputation.

However going public also entails costs. Some are straightforward: the underwriting fees charged by the investment banks that assist the company in listing and placement, stock exchange listing fees, and the costs of complying with disclosure requirements. Other costs are more subtle, involving the negative fallout of disclosure due to greater visibility to competitors and tax authorities. Since some of these costs have a large fixed cost component, it is not surprising that many small companies refrain from going public: indeed Pagano, Panetta, and Zingales (1998) report that company size is the single most important determinant of the probability of going public. Recently, public liquid markets appear to have lost some of their former appeal for firms, judging from the steep decline in the number of publicly listed U.S. firms since 1997 (Doidge et al. 2017). This may be due both to the phenomenal growth in the availability of private equity funding and to the increased cost of corporate disclosure (see Section 12.5.3).

Most companies that go public do so via domestic IPOs, in the country where they are headquartered. Yet they can still increase their market access further by cross-listing their shares on foreign exchanges. Less frequently, companies undertake their IPO directly in a foreign market or simultaneously at home and abroad—a so-called global IPO. The benefits and costs associated with cross-listing are similar but not identical to those associated with the first listing discussed above.

On the benefit side, cross-listing may increase liquidity and access to new investors; indeed, companies tend to cross-list in more liquid and larger exchanges (Pagano et al. 2001). However, in contrast to an IPO, a cross-listing creates an additional market for the same stock. Hence, in theory, it might reduce the liquidity of the home market, or fail to produce a very active foreign market for the stock. In other words, the fragmentation of trading induced by cross-listing may reduce total liquidity (see Section 7.2)—an issue that has been addressed by a number of empirical studies (see the references in Section 12.6).

The effect of cross-listing on the cost of equity capital appears to vary with company's country of origin: non-U.S. companies from countries with relatively poor shareholder protection tend to get positive stock price reactions when they announce that they will cross-list in the United States, while no such reaction is observed for cross-listings by U.S. companies (Karolyi 1998). Cross-listing also enhances increased visibility and reputation: actually, of 305 cross-listed European companies, 57% reported that the most important benefits of a foreign listing are increased visibility and prestige (Bancel and Mittoo 2001).

This suggests that some benefits of cross-listing stem specifically from the wide range of choices of where to cross-list, while for the original IPO, listing on the domestic exchange is the default option. Hence, cross-listing can be used strategically to “bond” the company to the rules of an investor-friendly foreign jurisdiction or to benefit from the prestige and visibility of an established stock exchange. Indeed, companies tend to cross-list in countries with better investor protection (Pagano et al. 2001)—a choice generally rewarded by the market via a lower cost of capital or equivalently a “cross-listing premium.” Consistent with the bonding hypothesis, the premium of companies that cross-list in the United States is positively correlated with measures of improvements in corporate governance (see the survey by Karolyi 2006 and the references in Section 12.6).

Another strategic benefit of cross-listing is to access investors who are best at evaluating its business plans or its technology. Hence, a cross-listing can be used to enhance the informativeness of stock prices to managers and thereby firm value, as is shown by Foucault and Gehrig (2008). For instance, a firm that plans to launch a product in the U.S. market knows that American investors are well placed to judge its potential. One way for the firm to obtain credible feedback from U.S. consumers is to cross-list in the United States, thus encouraging local investors to trade its stock and contribute to price discovery. U.S. investors may also have greater expertise in evaluating the business of high-tech firms, such as software or biotech companies, which have a particularly strong presence in the U.S. economy.

The evidence on cross-listings and foreign IPOs is consistent with the thesis that companies seek to list where investors are better positioned to value their

shares. Pagano, Röell, and Zechner (2002) document that the foreign sales of European companies that cross-listed between 1986 and 1997 tended to expand afterwards, and that U.S. exchanges attracted mainly cross-listings of high-tech and export-oriented companies. The same motivations appear to apply to most companies that choose to make their original IPO outside their domestic market. Caglio, Weiss Hanley, and Marietta-Westberg (2011) document that foreign IPOs tend to involve high-tech firms strongly oriented towards foreign markets, and originate from countries with low capital market development and poor disclosure requirements.

12.5.2 Designated Market Makers

As was explained in Section 12.2, companies whose shares are more liquid in secondary market trading fetch a higher price at the IPO stage (Ellul and Pagano 2006): the fact that investors appear to value post-IPO liquidity suggests that firms can benefit from policies that improve the liquidity of their stock. And in fact, IPO underwriting banks sometimes act as market makers to boost the new stock's liquidity in the months following the offering (see Ellis, Michaely, and O'Hara 2000). Firms pay their underwriter an additional fee for this service, which attests the fact that liquidity is particularly valuable at the IPO stage.

But firms can enter into this type of arrangement even after the IPO: they can—and often do—hire a market-maker to maintain a liquid market in their stock, by keeping the spread below some agreed level, for instance. Such designated market makers (DMMs) have been increasingly popular in LOB markets: they have appeared in France, Germany, Italy, the Netherlands, Sweden, and Norway, where they operate with aggressive limit orders on the buy and sell side of the book and thereby increase liquidity to incoming orders. In this sense, their operation is completely different from that of the NYSE specialists, whose last-mover advantage has such potential drawbacks, as the possibility of cream skimming, which might harm liquidity (Section 6.3.3).

Indeed, the evidence shows that DMMs have increased market liquidity—as measured by spreads and other indicators—in all the markets where they have been introduced (see Section 12.6 for the relevant references). The studies by Menkveld and Wang (2011) for the Netherlands and by Skjeltorp and Ødegaard (2015) for Norway also show that the introduction of a DMM is associated with a reduction in liquidity risk, as measured by the “liquidity betas” proposed by Acharya and Pedersen (2005), described in Chapter 10. These studies conclude that the introduction of a DMM corresponds to statistically and economically significant abnormal returns, which vary considerably across markets. The average yearly cumulative abnormal return associated with their

announcement ranges from 1% in Oslo to 3.5% in Amsterdam, 5% in Paris, and 7% in Stockholm. In any event, this is concurrent evidence that liquidity is valued by the market, well after the IPO stage.

Since hiring a DMM is a corporate decision, one wonders which companies choose to spend money to enhance market liquidity and what type of benefit is obtained. The study by Skjeltorp and Ødegaard (2015) on DMMs in the Oslo stock market is enlightening. It shows that only relatively illiquid firms choose to hire a DMM (see Figure 12.6). The histogram in panel A shows the frequency distribution of the relative bid-ask spread for the companies that do not have a DMM in a given year; panel B displays that information for firms that do hire a DMM, both the year before and the year after the hire. Comparing the three plots, we see that the most liquid firms do without a DMM—probably because they are confident that their bid-ask spreads are narrow anyway—and for companies that do hire a DMM, the distribution of spreads is distinctly shifted to the left.

This study also sheds light on why companies hire a DMM, showing that the decision is affected by the likelihood that the firm will interact with the capital markets in the future. Firms that choose to hire a DMM have greater growth

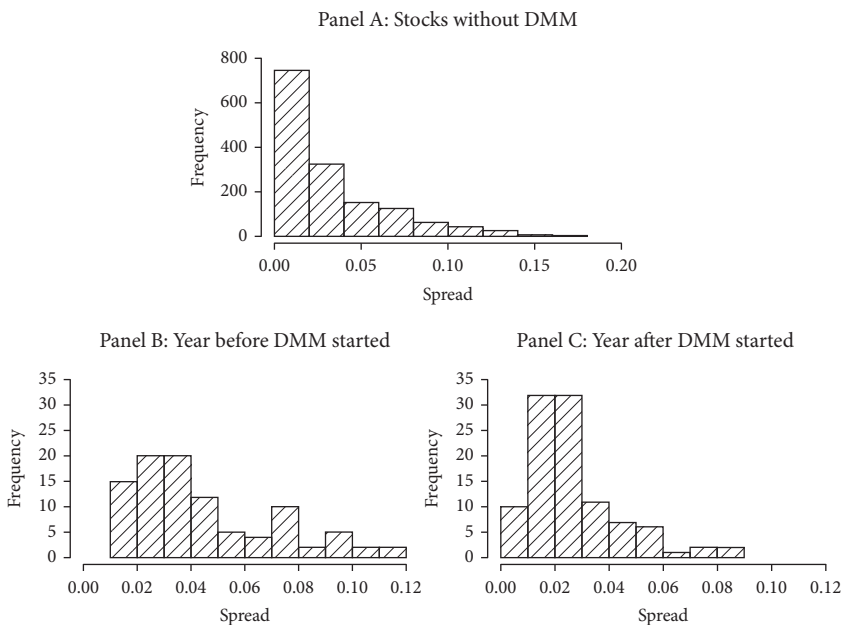


Figure 12.6 Frequency distribution of bid-ask spread for stocks without DMM and with DMM.

Source: Figure 2 in Skjeltorp and Ødegaard 2015.

opportunities (as measured by their Tobin's Q) and are more likely to raise external equity after the appointment of the DMM. Presumably they introduce a DMM precisely because they plan to issue more shares in the future and hope to do so at better prices.

12.5.3 Disclosure Policy

As observed in Section 12.5.1, committing to a high degree of corporate disclosure is part and parcel of the decision to go public. First, at the IPO stage a company must file a prospectus, which provides certified balance sheet data, as well as information about ownership and control, business plans and expected performance, reasons for going public, and so on. Second, from that moment onward, the company commits to periodic disclosure of certified balance-sheet data and regular earnings announcements. Third, blockholders commit to disclose information about trades that imply a substantial change in their stake in the company and on related-party transactions (e.g., deals with important suppliers or customers).

However, if the shareholders want, they can opt for even more disclosure than required by the domestic stock market or regulator. For instance, they can cross-list in a market with more stringent requirements or better enforcement against insider trading, or in a jurisdiction with stronger shareholder protection. All of these policies—see the evidence on the “bonding hypothesis” in Section 12.5.1—enable companies to secure a lower cost of capital. But cross-listing is not the only way to commit to better disclosure: a company can do so by having its accounts certified by reputable auditing companies, by adopting international rather than national accounting standards, or by committing to regular meetings with analysts to set out the company's prospects.

Insofar as they mitigate adverse selection problems, all these disclosure mechanisms should help firms to raise cheaper and more abundant equity capital not only in the IPO but also in subsequent seasoned equity offerings. Indeed, there is empirical evidence that firms that choose greater transparency tend to attract more funding from investors and have a lower cost of capital. Since greater public disclosure reduces the rents from private information, it should also increase market liquidity, which itself lowers the cost of capital, as shown in Chapter 10.

The liquidity-promoting effect of disclosure is documented by several studies: when firms switch to an accounting regime that features better disclosure, their stocks typically become more liquid. This effect is confirmed by cross-country evidence. Lang, Lins, and Maffett (2012) show that firms that choose a higher degree of transparency (as measured by less earnings management, better accounting standards, higher quality auditors, more analyst following, and more

accurate analyst forecasts) have narrower bid-ask spreads and fewer non-trading days. They also demonstrate that this increased liquidity is associated with lower cost of capital and higher valuation. Further evidence on the relationship between transparency, liquidity, and access to capital markets comes from what happens to companies when they lose all of their analysts' coverage (Figure 5.2). Their liquidity suffers compared to a control sample of similar companies (Ellul and Panayides 2018), and their investment drops significantly (Derrien and Kecskés 2013), suggesting that their access to capital diminishes or their cost of capital increases. To some extent, firms may compensate for the adverse effects of loss of coverage by analysts by disclosing more information, as found empirically by Balakrishnan et al. (2014). Section 12.6 reports further evidence on these issues.

If disclosure is so beneficial for firms, why is it that not all firms opt for maximum disclosure? The fact is that disclosure also has costs, and the trade-off between costs and benefits differs between companies and countries, depending on a variety of circumstances. The first, and most obvious, cost of transparency is the expense of disseminating the information in a credible way to investors (Section 12.5.1): listing fees, underwriting commissions, auditing fees, regulation compliance costs, and so on. For instance, compliance with the disclosure rules of a foreign jurisdiction sometimes requires switching to a different accounting system. Biddle and Saudagaran (1989) and Saudagaran and Biddle (1992) report that stringent disclosure requirements deter the listing of foreign companies, and the companies surveyed by Bancel and Mittoo (2001) place them among the chief disadvantages of a cross-listing.

A second, and less obvious, cost stems from the non-exclusive nature of information disclosure: once information about a firm's business plans or R&D projects is known to investors, it becomes known to competitors as well, who may exploit it to appropriate the firm's profit opportunities. This should be a particularly acute concern for high-tech, innovative firms—and especially the more successful ones: Campbell (1979) and Yosha (1995) suggest that firms with projects whose value depends on confidentiality may prefer bilateral financing arrangements rather than going public. Regulation may further exacerbate this concern. In the United States, the Regulation Fair Disclosure (known as Reg FD) prevents public companies from disclosing information selectively to certain groups of investors. The value of confidentiality for the increasing number of firms that rely heavily on patents and intangible capital may be one of the reasons why lately more and more firms prefer private equity financing to raising capital via a public listing.

A third, closely related cost arising from the non-exclusive nature of financial disclosure is diminished ability to evade or elude taxes: for instance, if the company provides more detailed accounting information to investors, it also

reveals it to the tax authorities, especially if it is incorporated in a country that prescribes “book-tax conformity” (i.e., that bans reporting different earnings to tax authorities and investors). So in choosing accounting transparency, firms must trade off the benefits of access to more abundant and cheaper capital against the cost of a greater tax burden. Ellul et al. (2018) find that cross-country company-level data are consistent with the existence of this trade-off. First, investment and access to finance are correlated positively with accounting transparency and negatively with tax pressure. Second, transparency is negatively correlated with tax pressure: firms choose less transparency where the implied tax cost is higher, particularly in sectors where they are less dependent on external finance, so that the implied cost in terms of foregone investment is lower.

Finally, disclosing more information forces investors to work harder to appraise the pricing implications, and these information processing costs—whether in terms of personal effort or of the expense on additional financial advice—must be compensated by higher returns. Worse, if some investors are more sophisticated than others (e.g., better trained at pricing securities), disclosing more detailed information about companies or securities may increase the informational advantage of sophisticated investors over unsophisticated ones. In this case, paradoxically, disclosure may *exacerbate*, rather than mitigate, adverse selection problems. In other terms, the thesis that transparency reduces information asymmetries rests on the important assumption that all investors are equally good at processing the additional information. Failing this assumption, disclosing more information may lead unsophisticated investors to leave the market or to require a larger return on their investment to overcome the expected losses from investing in markets where they are disadvantaged. Thus, issuers may prefer to provide only coarse information for complex securities when information is particularly costly for many investors to process. Pagano and Volpin (2012), who make this point with reference to the asset-backed securities (ABS) created by the securitization process, also observe that while opacity may reduce adverse selection at the issue stage, it creates the danger that adverse selection may reappear in secondary trading, if at that stage sophisticated investors are still able to dig out the undisclosed information.¹⁰

Some readers may wonder how the notion of “market transparency” extensively analyzed in Chapter 8 relates to the transparency that companies achieve

¹⁰ Another potentially harmful effect of disclosure is that when it refers to the ownership and control structure of the firm, it may discourage the accumulation of large blocks of shares and so hinder shareholder activism (Section 12.3). Of course, this effect is harmful for companies only insofar as shareholder activism is value-increasing.

via the disclosure policies discussed here. The two notions obviously refer to quite different “objects”: Chapter 8 refers to the market trading process—how much is known about past trades, incoming orders, and existing quotes on a company’s security. The disclosure policies considered here refer to a company’s fundamental value, insofar as this can be gauged from its accounts, business plans, ownership, control, and so on. Bearing this in mind, we may call this “fundamental transparency.” In any case, both forms of transparency, though different, can reduce adverse selection and increase liquidity, and both are therefore often rewarded by investors via higher issue prices or more generous funding. These two forms of transparency can be considered as substitutes, in the sense that an issuer who wishes to achieve a certain target level of liquidity and cost of capital may opt for greater corporate disclosure (more “fundamental transparency”) to compensate for the opacity of the trading process (less “market transparency”).

12.5.4 Capital Structure

Throughout this book the payoff of a stock has been taken to be a random variable v , equal to the fundamental value of the firm that issued the stock (standardizing for simplicity the number of shares to one). Hence, the rate of return offered by the company, $r = v/p - 1$, coincides with the return on its stock. This is appropriate if the firm is entirely equity financed. But if it has also issued some debt securities, the future cash flow v must be split between shareholders and debtholders in proportion to their holdings, and the return on equity will differ from the total return on the company’s assets. Denoting the initial value of the company’s equity by E and that of its debt by D , the total return on assets is the weighted average of the return to its equity and debt components:

$$r = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D.$$

Hence, the return on the company’s stock is:

$$r_E = \frac{E+D}{E}r - \frac{D}{E}r_D.$$

This expression shows that the more highly leveraged the firm, the greater the volatility of a shareholder’s rate of return: a 1% change in the firm’s total return r will determine a $\frac{E+D}{E}$ percent move in its stock return r_E , if the return on debt r_D stays constant. News about the firm’s future cash flow disproportionately affects the payoff to shareholders, because—if debt is and remains risk free—it leaves

the rate of return to debtholders unaffected. Hence, the riskiness of stocks is not determined only by the riskiness of fundamentals, but also by the firm's choice of its leverage ratio D/E : more generally, capital structure determines how the final value is split among different securities, such as equity, debt, hybrids, and options, and therefore the stochastic characteristics of the returns to each of these securities.

This has an important implication for liquidity. If some investors have better access to news about fundamentals (or are better at processing it), the asymmetry of information will create illiquidity only for information-sensitive securities, such as stocks, and not for safe ones, such as debt. Gorton and Pennacchi (1990) show that firms can mitigate the trading losses associated with informational asymmetry by designing securities with cash flows that are insensitive to private information and so can be safely bought by uninformed investors who want them only for their liquidity needs. Firms have an incentive to issue such safe securities, because they will be very liquid and, therefore, *ex ante* their buyers require no liquidity premium to hold them. Informed investors will instead concentrate their trades on information-sensitive securities, because these are the only ones from which they can hope to earn informational rents.

Of course, if the investment needs of uninformed investors are completely satisfied by the issuance of safe corporate debt by companies, then there are no stock market rents to be obtained by informed investors, because such rents arise from the trading losses incurred by the uninformed. If uninformed investors' wealth exceeds the amount of safe debt that companies can issue, in equilibrium they must hold and possibly trade some stocks. And the more leveraged the company, the more information sensitive and therefore the more illiquid these stocks will be. In this case, even the issuance of corporate debt does not completely shelter uninformed investors from the need to hold some illiquid securities and pay some transaction costs. This will restore some rents for informed investors and make stocks attractive for them.

However, as Section 12.4.1 points out, the illiquidity of stocks comes with a potentially countervailing benefit, namely, the improved price discovery implied by informed trading. And, since greater leverage implies that stocks are more information-sensitive, by increasing their leverage firms can increase the attractiveness of their stock to informed traders. If the proportion π of informed traders is not constant, but varies with informational rents, greater leverage will translate into a higher value of π , hence both higher trading costs for uninformed investors (equation (12.12)) and greater overall firm value (equation 12.7).

This suggests that in choosing its degree of leverage, and therefore the extent to which it wants to attract informed trading, a firm may seek to balance the higher trading costs that it thereby inflicts on its uninformed investors—and

thus the illiquidity premium that it will have to pay, as argued in Chapter 9—against the benefits of better price discovery. The extent to which a company wishes to leverage and thus attract informed trading will be all the more limited if uninformed traders' participation in the market is sensitive to the bid-ask spread: in response to more informed trading, they may curtail or abandon their trading altogether and thus reduce the profits available to informed traders, which by equations (12.10) and (12.11), are increasing in $1 - \pi$. Thus, information collection will be discouraged and there is a limit to how high π , and with it, stock price informativeness, can be made.

The trade-off between liquidity and the informativeness of stock prices is also affected by the value of information for firms' decisions: Chang and Yu (2010) argue that when information is not very important to improving operating decisions, a firm should limit its debt issuance, thereby reducing the information-sensitivity of its stock price. In this way, the firm discourages acquisition of private information, which reduces the illiquidity premium associated with adverse selection. This may contribute to explain why many firms issue no or modest amounts of debt, which is one of the main puzzles in corporate finance, given the compelling tax advantages of leveraging up to the hilt.

12.6 Further Reading

Stock market development, investment and growth. There is a vast literature on the relationship between financial market development, investment, and growth, which is surveyed by Pagano (1993a), Demirguc-Kunt and Levine (2001), and Levine (2005), among others. Besides Henry (2000b), other studies of the effects of stock market liberalization are Bekaert, Harvey, and Lundblad (2005), who show that equity market liberalization is associated with a subsequent average annual real economic growth of about 1%, and that this effect is robust to controls for capital account liberalization and other simultaneous reforms, and Gupta and Yuan (2009), who document industries that are more externally dependent and face better growth opportunities grow faster following liberalization. Moreover, Levine and Zervos (1998b) find that such policies are associated with an increase in stock market liquidity, and Henry (2000a) and Bekaert and Harvey (2000) document that they coincide with a jump in stock prices and a drop in the cost of equity capital.

Informational effects of market prices on firms. Bond, Edmans, and Goldstein (2012) offer a good survey of the literature on the real effects that financial markets have on firms' real decisions via the information provided by security prices. Dow and Gorton (1997) and Subrahmanyam and Titman (1999) provide early theoretical analyses of the feedback from asset prices to firms'

investment decisions, via managerial learning. Testing for this channel has subsequently attracted a lot of attention by empirical researchers. Using U.S. data, Chen, Goldstein, and Jiang (2007) find that firms' investment decisions respond positively to their stock price, consistent with the thesis that managers gather information from stock prices. Moreover, in accordance with the model presented in Section 12.4.1, the sensitivity of investment to stock prices is greater for firms with more informative stock prices: firms that attract more informed investors (as measured by the PIN variable presented in Chapter 5) feature greater sensitivity of investment to prices. A non-exhaustive list of other empirical studies on this issue includes Durnev, Morck, and Yeung (2004), Luo (2005), Bakke and Whited (2010), Foucault and Frésard (2012), Foucault and Frésard (2014), or Dessaint, Foucault, Frésard, and Matray (2019).

Shareholder activism. Edmans (2014) provides a survey of the literature on blockholders and corporate governance. Several authors have analyzed the positive externality created by a large shareholder's monitoring, and the implied holdout problem. Shleifer and Vishny (1986) show that if a takeover bidder starts with zero initial holdings, he has no incentive to acquire shares and become an activist shareholder. Huddart (1993) and Admati, Pfleiderer, and Zechner (1994) analyze this externality in models where the potentially activist shareholder is risk-averse. Admati and Pfleiderer (2009) show that exit and voice are not always mutually exclusive, since a large shareholder may discipline managers by threatening exit on the basis of private information. Stoughton and Zechner (1997) and Pagano and Röell (1998) analyze models in which the stake of the activist blockholder is determined endogenously at the IPO stage. On the empirical front, Parrino, Sias, and Starks (2003) document the preference for exit over voice by U.S. institutional investors.

Managerial compensation and incentives. The literature on executive compensation is vast. A good number of models have shown that performance-based compensation in the form of stock and options can enhance the incentive to exert effort and take risk (see, for instance, Smith and Stulz 1985; Hall and Murphy 2000; and Dittmann and Maug 2007). But more recent work highlights the possible distortionary effects of such managerial incentives. Theoretical work by Goldman and Slezak (2006) and Peng and Röell (2008b, 2014) focuses on the efficiency costs of manipulation, and empirical work by Bergstresser and Philippon (2006), Burns and Kedia (2006), Kedia and Philippon (2009), and Peng and Röell (2008a) demonstrates that high-powered incentive schemes (especially options) are positively correlated with proxies for accounting fraud, such as discretionary accruals, fraud accusations, accounting restatements, and class action litigation. Another distortionary effect of performance-based compensation is that it may lead to excessive risk taking by financial intermediaries and thus contribute to financial instability: indeed there is evidence of a positive

association between performance-based compensation and bank risk taking in the context of the 2008–9 financial crisis (for instance, Boyallian and Ruiz-Verdú 2018 and Kolasinski and Yang 2018).

Cross-listing. Karolyi (1998) surveys the evidence on stock price behavior around cross-listings and finds that the effect differs across companies; even when initially positive, the effect is often dissipated in the subsequent year. On balance, non-U.S. companies listing in the United States have positive cumulative abnormal returns (Foerster and Karolyi 1999) and experience a reduction in the cost of capital (Karolyi 1998). Foerster and Karolyi (1999) show that the stock prices of cross-listed companies rise more significantly when the expansion of the shareholder base is more pronounced. Foerster and Karolyi (1999), Kadlec and McConnell (1994), and Miller (1999) show that the price reaction to a cross-listing is positively correlated with the increase in the shareholder base.

Several studies document that cross-listing is associated with an increase in liquidity, in the form of lower bid-ask spreads and more trading: see Kadlec and McConnell (1994); Noronha, Sarin, and Saudagaran (1996); Foerster and Karolyi (1998, 2000); and Smith and Sofianos (1997). However, cross-listing may not always enhance liquidity, due to the potentially offsetting impact of market fragmentation, as in the models of Pagano (1989), Chowdhry and Nanda (1991), and Madhavan (1995). Indeed the evidence on the emerging markets indicates that cross-listing in the United States tends to depress domestic trading. Domowitz, Glen, and Madhavan (1998) show that the home market liquidity of Mexican companies decreases upon their issuance of American depository receipts; they relate this effect to the poor information linkages between the two markets. This is consistent with Levine and Schmukler (2006), who find that companies in emerging economies experience a drop in domestic trading when they cross-list, and with Halling et al. (2008), who document that a cross-listing has a positive effect on home market liquidity for companies in developed countries, but a negative one in emerging countries.

On the bonding hypothesis, Doidge (2004) ties the premium from cross-listing in the United States to the cross-listing's ability to reduce the voting premium, and therefore produces direct evidence on its effect on the controlling shareholders' ability to extract private benefits and increase the firm's ability to take advantage of growth opportunities. Doidge, Karolyi, and Stulz (2004) document that in 1997 foreign companies listed in the United States had a Tobin's *Q* ratio 16.5% higher than that of firms from the same country not listed in the United States. This difference is statistically significant and persists even after controlling for several firm and country characteristics. They also find that growth opportunities are valued more highly for firms listed in the United States especially for firms from countries with less investor protection. Doidge, Karolyi, and Stulz (2009) show that in spite of the greater compliance costs associated

with the Sarbanes-Oxley Act in the United States, the listing premium associated with a New York listing persists, while no such premium is associated with a London listing.

Designated market makers (DMM). Nimalendran and Petrella (2003) show that the introduction of a specialist-based market-making mechanism for thinly traded Italian Stock Exchange securities was associated with a reduction in execution costs. Venkataraman and Waisburd (2007) find that the introduction of a DMM by relatively illiquid liquid firms listed on the Paris Bourse improves market quality and that its announcement is associated with a 5% cumulative abnormal return. They also show that DMM are mainly hired by younger and smaller firms and by those with less volatile stocks. Hengelbrock (2008) studies the effect of the introduction of DMM for thinly traded stocks on the Xetra platform of the Deutsche Börse, and finds that while quoted and effective spreads narrow when trading with one or two designated sponsors, further increases in the number of specialists do not necessarily result in greater liquidity. Anand, Tanggaard, and Weaver (2009) show that DMM introduction in Sweden increases liquidity, produces a 7% cumulative abnormal price run-up, increases volume, and leads DMMs to trade more in the stocks that they contract for. Menkveld and Wang (2011) study the introduction of DMM in the Amsterdam stock market, when Euronext allowed listed firms to hire DMMs on October 29, 2001, and find that for 74 stocks that hired a DMM, liquidity increased, liquidity risk dropped, and there was an average cumulative abnormal return of 3.5%. Interestingly, DMMs participate in more trades and incur a trading loss on high quoted-spread days (days when their constraint is likely to bind). Finally, Bessembinder, Hao, and Zheng (2015) show that firms can benefit from hiring a DMM as this is a way to make the liquidity for their stock higher than it would be in a competitive market.

Transparency, liquidity, and access to capital. Leuz and Wysocki (2008) survey empirical work on the relationship between transparency and firms' access to capital markets. This literature consistently shows that increased disclosure is associated with greater stock market liquidity and lower cost of capital. Goldstein and Yang (2017) provide a survey of the relevant theoretical literature.

The idea that information disclosure may exacerbate adverse selection problems, discussed in the text with reference to Pagano and Volpin (2012), is also present in the theoretical work by Dang, Gorton, and Holmstrom (2010) and Morris and Shin (2012). In the same line of argument, Kim and Verrecchia (1994) show that earnings announcements decrease market liquidity if they allow sophisticated traders to increase their informational advantage over other traders. The same argument is used by Goel and Thakor (2003) to rationalize earnings smoothing: to maintain a liquid market for their stocks, companies

will smooth earnings so as to reduce the informational rents of sophisticated investors. Di Maggio and Pagano (2011) analyze the relationship between financial disclosure (“fundamental transparency”) and “market transparency” in a model where investors have different information processing costs and securities are sold via a search market.

Empirically, the practice of selling securities exclusively in bundles, rather than separately, may be considered as one instance in which opacity increases liquidity: Kavajecz and Keim (2005) show that asset managers lower trading costs by 48% via “blind auctions” of stocks, whereby they auction a set of trades as a package to potential liquidity providers, without revealing the identities of the securities in the package to the bidders. Similarly, Vickery and Wright (2010) document that mortgage-backed securities issued by U.S. public agencies (agency MBS) are extremely opaque when placed via the “to-be-announced” market, where MBS sellers specify only a few basic characteristics of the security to be delivered, but are extremely liquid.

Capital structure, liquidity, and stock price informativeness. Gorton and Pennacchi (1990) argue that information-insensitive securities such as debt are more liquid. This thesis is also central to the model of Dang, Gorton, and Holmstrom (2010). The idea that increasing the information sensitivity of the firm’s securities can be beneficial because it elicits information collection by sophisticated investors is formalized by Boot and Thakor (1993) and Fulghieri and Lukin (2001).

12.7 Exercises

1. Trading disclosure and shareholder activism. Consider the blockholder’s decision whether to intervene or not in the company’s management at date 1 in the model in Section 12.3, and suppose that regulation forces him to disclose his intention to exit before trading, so that if he sells the market anticipates that no improvement will occur ($\mu_1 = V$). Show that in this case activism is guaranteed: the probability of intervention is $\pi = 1$.
2. **Stock market as guide to investment.** In the model analyzed in section 12.4.1, we assumed that $G < I$ and found that on average the stock market encourages investment. Consider here the complementary case where $G > I$:
 - a. Show that in this case the stock market deters inappropriate investment. Explain the intuition behind this result.
 - b. Assume that the market is informative enough for the stock price to affect investment and derive the new expression for the increase in firm’s ex ante value resulting from the presence of a stock market.

3. **No price manipulation by speculators.** In the context of the model of Section 12.4.1, verify that in equilibrium the informed speculator does not have an incentive to trade against his information, that is, to sell on good news or buy on bad news. Assume that the informativeness condition (12.5) holds.
4. **Bid and ask prices when investment does not react to stock prices.** In the model analyzed in Section 12.4.1, assume that the informativeness condition (12.5) does not hold.
 - a. Compute the equilibrium bid and ask prices, and show that in this case too, in equilibrium the informed speculator will buy upon receiving positive information and sell otherwise.
 - b. Compute the bid-ask spread and find out whether in this case it is increasing in the frequency of informed trading π .
 - c. Compare this bid-ask spread with that given by equation (12.12) under the informativeness condition (12.5).
 - d. Finally, compute the midprice and see whether it depends on the frequency of informed trading π , explaining why.
5. **Bid and ask prices when investment must be chosen before stock trading.** Consider the following change in the model analyzed in Section 12.4.1: assume that, even though the informativeness condition (12.5) holds, the manager cannot react to the stock price because stock trading takes place once the investment decision has already been irrevocably made. (Technically, we are reversing the order of the trading and investment stages of the game along the time line.) Assuming that the manager's investment decision is private and that market participants cannot observe it before trading, derive the ask and bid prices under this new assumption, and establish whether the market is more or less liquid than in the model of Section 12.4.1, by comparing the bid-ask spread that you have computed with that of equation (12.12). Explain the intuition behind your finding.
6. **Incentive pay when stock price is imperfectly informative.** Consider an extension of the model of Section 12.4.2 in which the stock market is informed (i.e., able to observe the manager's effort) only with frequency π , and is uninformed with the complementary probability $1 - \pi$.
 - a. Show that if shareholders write an incentive contract conditional on the stock price only, the compensation that they will have to pay is inversely related to π (the proportion of time that the stock market observes effort). [Hint: in your analysis you may assume that the stock market

rationality expects the manager to exert effort if his pay contract respects his incentive constraint.]

- b. (Harder problem.) Assuming that $\pi < 1$, investigate whether shareholders can obtain a better outcome by conditioning pay $w(P, v)$ on *both* the interim stock price *and* the final value of the firm. What would the pay contract look like?

